Warren Kozak

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Education

Carleton College — September 2023 – June 2027
Bachelor of Arts in Computer Science, Mathematics; Cumulative GPA: 3.9/4.0 — *Northfield, MN*

- Relevant Coursework: Math of Computer Science, Data Structures, Algorithms, Advanced Software Design, Programming Languages: Design and Implementation, Introduction to Computer Systems, Linear Algebra, Ordinary Differential Equations, Abstract Algebra, Robotics, Computability and Complexity

St. John's School — September 2019 – May 2023
High School Diploma; GPA: 95.13/100 — *Houston, TX*

- SAT: 1540 (EBRW: 780, Math: 760)

Experience

Northwestern University — July 2025 – August 2025
Student Researcher — *Evanston, IL*

- Prototyped wearable devices for use in swimming using a Seeed Studio nRF52840 Sense
- Optimized data logging with FreeRTOS, increasing IMU sample rate from 25 to 50 Hz for more reliable data
- Utilized a Mahony filter algorithm to generate quaternions from 6-axis IMU data to detect device orientation
- Developed an LSTM model to classify IMU data into stroke categories, achieving an accuracy of 95%

Carleton College — September 2024 – June 2025
Digital Scholarship Intern — *Northfield, MN*

- Prevented API rate-limiting errors by implementing a database caching solution for OpenRouteService
- Developed new features in PHP for Wordpress and Omeka plugins maintained by Carleton College
- Researched current digital humanities projects and led discussion groups about the role of DH at Carleton

Hildebrand Technology — June 2024 – August 2024
Summer Intern — *London, UK*

- Developed a Home Assistant integration in Python supporting Hildebrand Glow IoT devices via MQTT
- Integrated Glow data stream with the UK National Grid API to deliver actionable insights to users
- Authored comprehensive documentation to guide users through the integration setup process

Projects

codeduel.io | *Python, JavaScript, Flask, PostgreSQL, Docker* — July 2025

- Achieved second place out of 500+ submissions in the 2025 Boot.dev Hackathon (Professional Category)
- Worked with partner to build online game where users compete by completing coding challenges
- Built server side logic to execute submitted code in a secure container with minimal latency (< 5 seconds)

Scheme Interpreter | *C, Scheme, Makefile* — April 2025 – June 2025

- Implemented a 1000-line Guile Scheme interpreter in C consisting of a tokenizer, parser, and interpreter, supporting recursion, closures, special forms (if, let, define), and primitives (map, cons, car, cdr)
- Developed a custom implementation of talloc, achieving 0 memory leaks under Valgrind

WOS | *C, Assembly, Nix, Makefile* — January 2022 – May 2022

- Designed and implemented a simple kernel in C and Assembly with IRQ support
- Developed a timer and keyboard subsystem to take user keyboard input and display system uptime

Activities

Men's Varsity Swim & Dive | *Carleton College* — September 2023 – Present

- Swam at 2025 MIAC Championships, achieving all-MIAC in the 200 medley and freestyle relays
- Elected by teammates to serve as captain for the 2026-2027 season

Technical Skills

Languages: C/C++, Java, Python, Lisp/Scheme, SQL, LaTeX, JavaScript
Frameworks: ROS2, FreeRTOS, Arduino, Flask, React
Developer Tools: UNIX/Linux, Git, PlatformIO, Github, Docker, Neovim, VS Code, Eclipse
Libraries: Keras, TensorFlow, pandas, NumPy, SciPy, OpenCV, NLTK, Matplotlib